

LABHYA MEHTA

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EDUCATION

Cornell University

Expected Graduation:
May 2028

B.S. Mechanical Engineering

Minor: Aerospace Engineering,
Computer Science

COURSEWORK

- Dynamics*
 - Mechanical Synthesis*
 - Linear Algebra*
 - Waves and Oscillations*
 - OOP & Data Structures*
 - Engineering Simulations & Design*
 - Statics & Mechanics of Solids
 - Electromagnetism
 - Differential Equations
 - Multivariable Calculus
 - iOS Development
 - Mechanics and Heat
- *Planned Spring 2026

SKILLS

Design

CAD (SolidWorks, Creo), FEA & Simulation (ANSYS, XFLR5), Programming (Python, Java)

Manufacturing

Composite Manufacturing, Additive Manufacturing, Machining (Mill, Lathe)

EXPERIENCE

Mechanical Design Intern | Skygear

May 2025 – Aug 2025 | Gurugram, India

Skygear designs, develops, and manufactures capabilities for medium-altitude long-endurance (MALE) UAVs.

- Designed and fabricated a static engine test stand for a 110cc horizontally opposed internal combustion engine, incorporating load-bearing fixtures, vibration isolation, and throttle actuation
- Supported propulsion system integration on the Dronocopter UAV platform, assisting with engine mounting, CG alignment, and vibration isolation using elastomeric bushings
- Created 3D models & 2D manufacturing drawings in Creo for custom brackets, couplings, and support structures, and collaborated with technicians during fabrication and assembly
- Assisted in dynamic balancing of propeller–engine assemblies to reduce vibration, referencing applicable tolerance guidelines

Airframe Engineer | Cornell University Unmanned Air Systems

CUAir is a student-run team that designs, manufactures, and programs a custom eVTOL UAV.

Nov 2024 – Present | Ithaca, NY

- Led the structural design of an inverted V-tail, including spars, spar caps, ribs, and structural versus non-structural skin; extracted flight loads from XFLR5 and translated them into internal load paths and component placement using hand calculations
- Performed stress and deflection analysis of tail spars and spar caps in ANSYS; iterated geometry and layups to meet stiffness and strength requirements under flight loading.
- Modelled ribs, spars, spar caps, connectors, and rib supports in SolidWorks; fabricated ribs and structural supports via laser cutting and FDM 3D printing for airframe integration
- Designed aerodynamic boom caps and pitot tube mounts to improve aerodynamic profiling and structural packaging for avionics sensors
- Manufactured wings, tail, fuselage, and structural skeleton of a 12-ft wingspan eVTOL aircraft using a carbon fibre, fibreglass, and epoxy resin composite wet layup process

Undergraduate Researcher | Cornell Aerospace Adversary Laboratory

Cornell Aerospace Adversary Lab focuses on developing and analysing threat-representative aerospace systems for testing and validating autonomous defense technologies.

Nov 2024 – Present | Ithaca, NY

- Designed and prototyped custom drone parts, including lightweight structural parts and streamlined external components, using SolidWorks and AutoCAD
- Tested and evaluated payload mounting systems for secure attachment and impact resistance under simulated takeoff and landing conditions
- Worked under Dr. Gregory Falco and collaborated with a cross-disciplinary team of engineers to design and refine UAV components integrating avionics and tracking systems